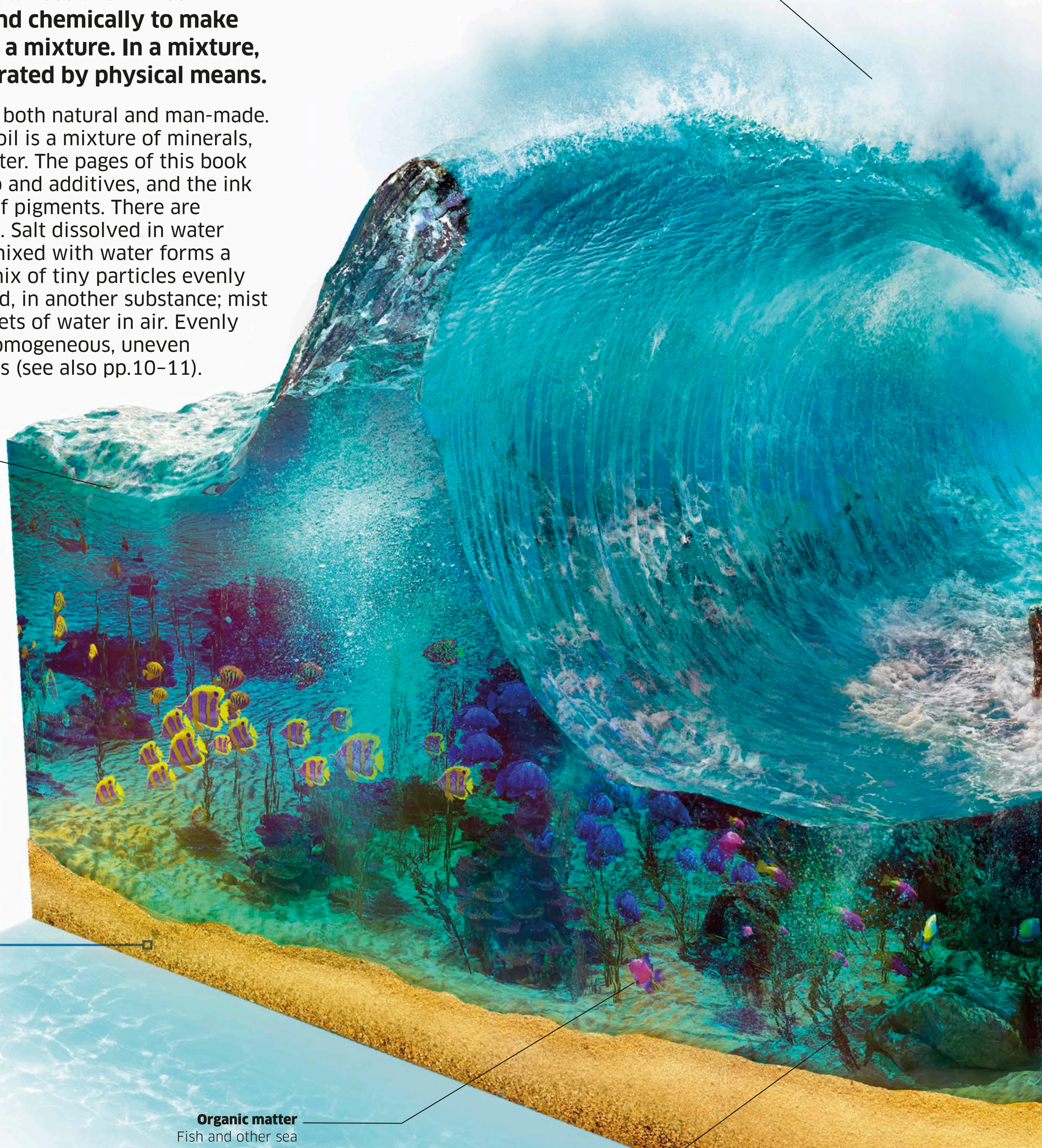
Ice cream is a **colloid mixture** of fats, air, ice crystals, and other substances.

Mixtures

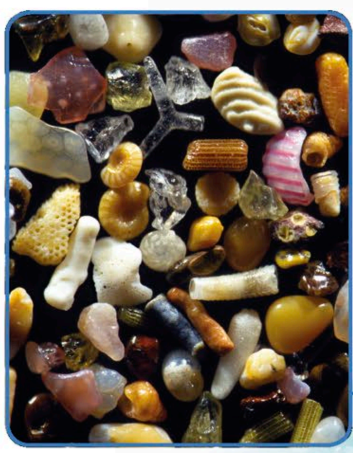
When two or more substances are mixed together, but do not bond chemically to make a compound, they form a mixture. In a mixture, substances can be separated by physical means.

Mixtures are all around us, both natural and man-made. Air is a mixture of gases. Soil is a mixture of minerals, biological material, and water. The pages of this book are a mixture of wood pulp and additives, and the ink on the pages is a mixture of pigments. There are different types of mixtures. Salt dissolved in water is a solution. Grainy sand mixed with water forms a suspension. A colloid is a mix of tiny particles evenly dispersed, but not dissolved, in another substance; mist is a colloid of minute droplets of water in air. Evenly distributed mixtures are homogeneous, uneven mixtures are heterogeneous (see also pp.10–11).



Spray
Sea spray is a heterogeneous mixture of air and seawater.

Salty solution
The salt water in the sea is a solution: a homogeneous mixture of water and dissolved salts. When seawater evaporates, salt crystals are formed.



Sand
Sand is a heterogeneous mixture: a close look reveals tiny pieces of eroded rock, crushed shells, glass, and even bits of plastic.

Organic matter
Fish and other sea creatures release organic matter, such as waste and old scales, into the sea.

Seaweed
Dead and decaying algae also contribute organic matter to the seawater mix.

Mixtures in nature

Most substances in nature are mixtures, including seawater, rocks, soil, and air. Understanding how to separate these mixtures provides us with an important supply of natural resources, for example by removing salt from seawater and separating gases, such as argon, from air.